

Mesh-dependent iteration count growth in primal-dual active set strategies

John Papadopoulos¹, Michael Hintermüller^{2,3}

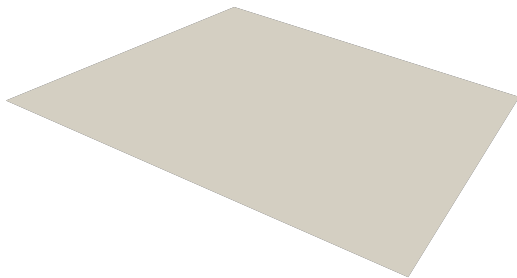
¹Oxford, ²Weierstrass Institute, ³Humboldt-Universität zu Berlin



CIGMO Seminar, 5 October 2026

Unconstrained infinite-dimensional optimization

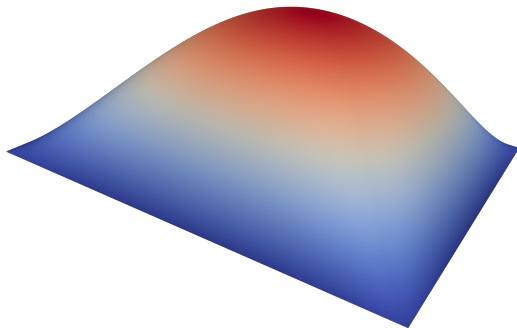
$$\min_{u \in H_0^1(\Omega)} \int_{\Omega} \frac{1}{2} |\nabla u|^2 - fu \, dx \Leftrightarrow \text{solve } -\Delta u = f \text{ (a PDE)}.$$



Minimize Dirichlet-type energy $\Leftrightarrow$ solve the Poisson PDE.

Unconstrained infinite-dimensional optimization

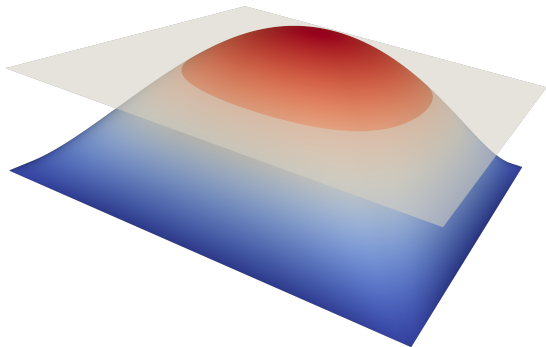
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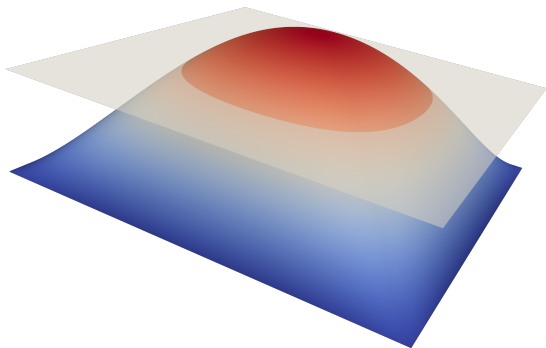
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Minimize energy with pointwise constraints $\Leftrightarrow$ solve the obstacle problem.

Pointwise constrained infinite-dimensional optimization

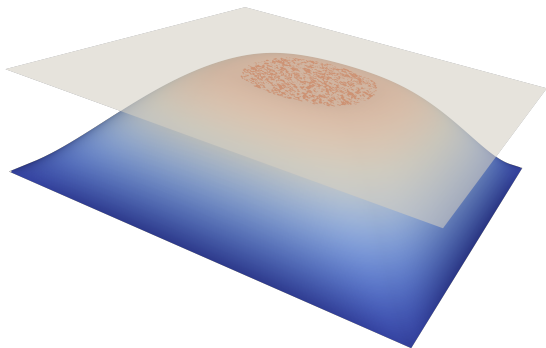
$$\min_{\substack{u \in H_0^1(\Omega) \\ u \leq \varphi}} \int_{\Omega} \frac{1}{2} |\nabla u|^2 - fu \, dx \Leftrightarrow \text{solve (inf-dim) KKT conditions.}$$



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Pointwise constrained infinite-dimensional optimization

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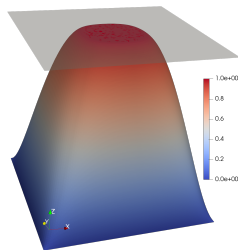
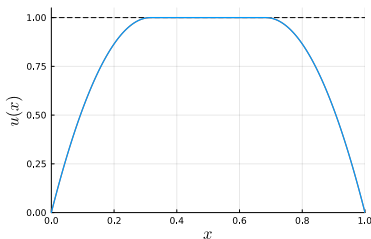
Minimize energy with pointwise constraints $\Leftrightarrow$ solve the obstacle problem.

Obstacle Problem

Problem Statement

Given f , φ , find $u : \Omega \rightarrow \mathbb{R}$, minimizing:

$$\min_{u \in H_0^1(\Omega)} \int_{\Omega} \frac{1}{2} |\nabla u|^2 - f u \, dx \quad \text{subject to } u \leq \varphi \text{ a.e. in } \Omega.$$



Obstacle problem with $\Omega = (0, 1)^d$, $d \in \{1, 2\}$, $f \equiv 20$, $\varphi \equiv 1$.

Obstacle Problem

P1-FEM discretization for u , f , and φ .

Discretized Obstacle Problem

Given $\mathbf{b}$, φ , find $\mathbf{u} \in \mathbb{R}^N$ minimizing:

$$\min_{\mathbf{u} \in \mathbb{R}^N} \frac{1}{2} \mathbf{u}^\top A \mathbf{u} - \mathbf{b}^\top \mathbf{u} \text{ subject to } \mathbf{u} \leq \varphi,$$

where $\mathbf{u} \leq \varphi \iff \mathbf{u}_i \leq \varphi_i$ for all $i \in \{1:N\}$.

A locally fast-converging solver

How do we solve the discretized obstacle problem?

A locally fast-converging solver

How do we solve the discretized obstacle problem?

What is a locally fast-converging solver for the obstacle problem?

For the **obstacle problem**, a classic locally fast-converging method is the **primal-dual active-set (PDAS) method**. It can also be interpreted as a **semismooth Newton method**, which explains its fast local convergence.

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Does PETSc have such a solver implemented?

Yes. In fact, PETSc has a solver very close to the **primal-dual active-set / semismooth Newton approach for obstacle problems**.

The most relevant PETSc solver is `SNESVINEWTONRSL` :

Hintermüller, Ito, Kunisch (2002)

The Primal-Dual Active Set Strategy as a Semismooth Newton Method,
SIAM Journal on Optimization,
<https://doi.org/10.1137/S1052623401383558>.

- ✓ Finite-dimensional quadratic programming problems.
- ✓ Infinite-dimensional optimal control with control pointwise constraints. Mesh-independent iteration counts.
- ✗ [Page 14] *At this point we remark that the primal-dual active set strategy has no straightforward infinite dimensional analogue for [...] obstacle problems [...] as the Lagrange multiplier is only a measure in general, and hence the core steps [...] of our algorithm are no longer meaningful.*

But we always discretize first anyway! So does it matter?

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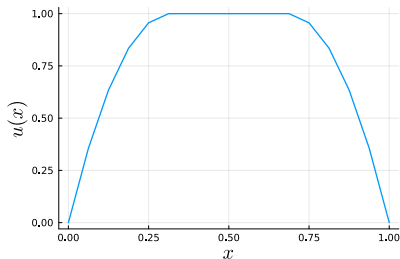
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Mesh-dependent iteration counts

Pick $\Omega = (0, 1)^d$, $\varphi \equiv 1$, $f \equiv 20$.

h	2^{-4}	2^{-5}	2^{-6}	2^{-7}	2^{-8}	2^{-9}	2^{-10}
$d = 1$							

PDAS iterations to reach convergence for on a sequence of uniform mesh refinements when $u^0 = 0$.

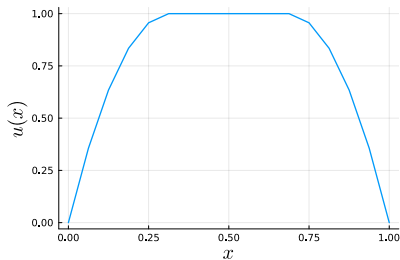
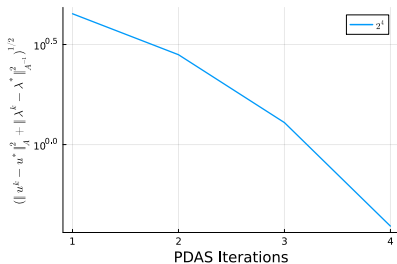


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Pick $\Omega = (0, 1)^d$, $\varphi \equiv 1$, $f \equiv 20$.

h	2^{-4}	2^{-5}	2^{-6}	2^{-7}	2^{-8}	2^{-9}	2^{-10}
$d = 1$	5						

PDAS iterations to reach convergence for on a sequence of uniform mesh refinements when $u^0 = 0$.

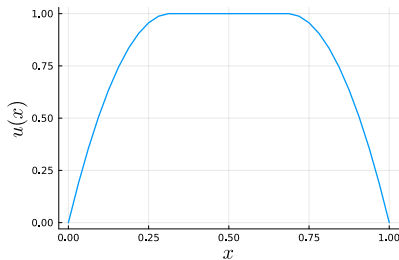
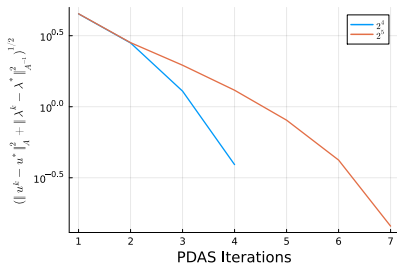


Mesh-dependent iteration counts

Pick $\Omega = (0, 1)^d$, $\varphi \equiv 1$, $f \equiv 20$.

h	2^{-4}	2^{-5}	2^{-6}	2^{-7}	2^{-8}	2^{-9}	2^{-10}
$d = 1$	5	8					

PDAS iterations to reach convergence for on a sequence of uniform mesh refinements when $u^0 = 0$.

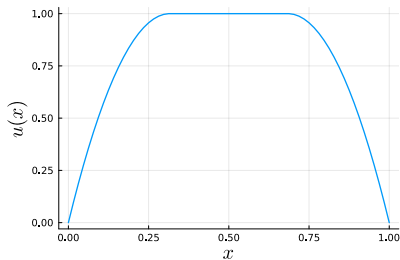
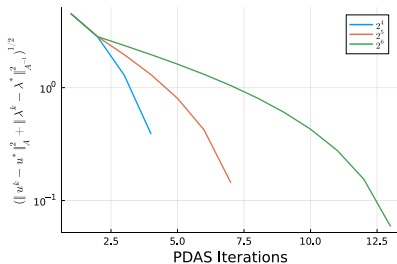


Mesh-dependent iteration counts

Pick $\Omega = (0, 1)^d$, $\varphi \equiv 1$, $f \equiv 20$.

h	2^{-4}	2^{-5}	2^{-6}	2^{-7}	2^{-8}	2^{-9}	2^{-10}
$d = 1$	5	8	14				

PDAS iterations to reach convergence for on a sequence of uniform mesh refinements when $u^0 = 0$.

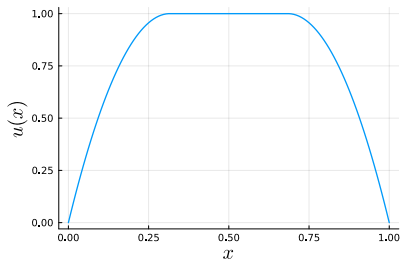
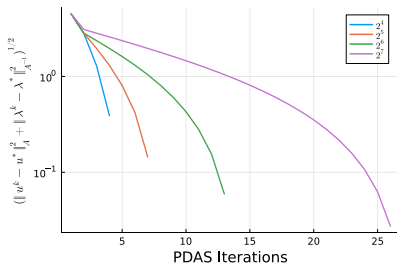


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h	2^{-4}	2^{-5}	2^{-6}	2^{-7}	2^{-8}	2^{-9}	2^{-10}
$d = 1$	5	8	14	27			

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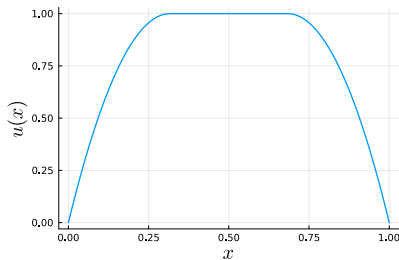
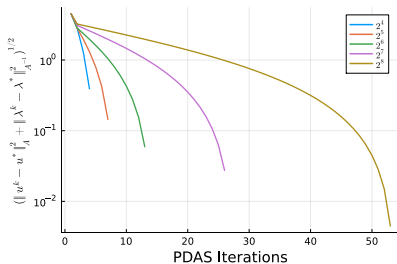


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Pick $\Omega = (0, 1)^d$, $\varphi \equiv 1$, $f \equiv 20$.

h	2^{-4}	2^{-5}	2^{-6}	2^{-7}	2^{-8}	2^{-9}	2^{-10}
$d = 1$	5	8	14	27	54		

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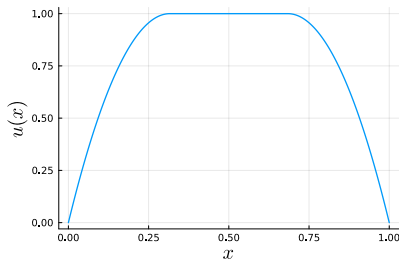
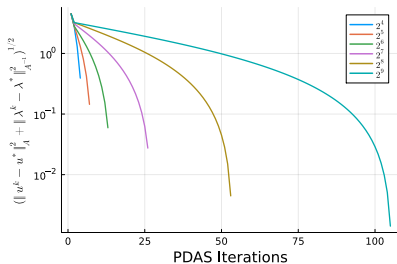


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h	2^{-4}	2^{-5}	2^{-6}	2^{-7}	2^{-8}	2^{-9}	2^{-10}
$d = 1$	5	8	14	27	54	106	

PDAS iterations to reach convergence for on a sequence of uniform mesh refinements when $u^0 = 0$.

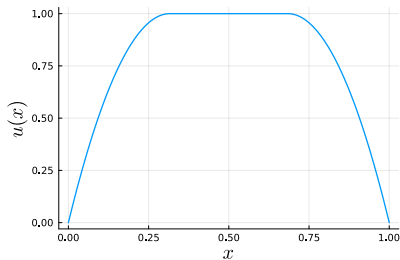
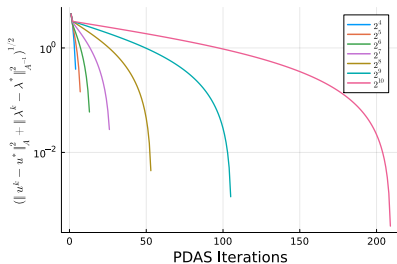


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Pick $\Omega = (0, 1)^d$, $\varphi \equiv 1$, $f \equiv 20$.

h	2^{-4}	2^{-5}	2^{-6}	2^{-7}	2^{-8}	2^{-9}	2^{-10}
$d = 1$	5	8	14	27	54	106	210

PDAS iterations to reach convergence for on a sequence of uniform mesh refinements when $u^0 = 0$.

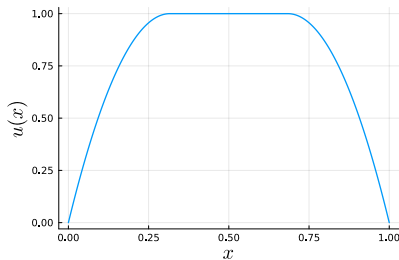
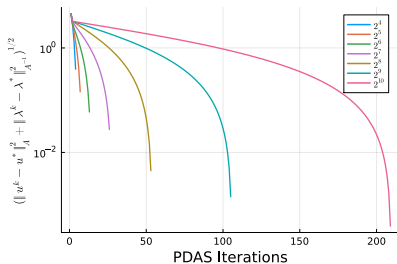


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$d = 1$	5	8	14	27	54	106	210

PDAS iterations to reach convergence for on a sequence of uniform mesh refinements when $u^0 = 0$.



The number of iterations to reach convergence roughly doubles with each uniform refinement!

Mesh-dependent iteration counts

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PDAS iterations to reach convergence for on a sequence of uniform mesh refinements when $u^0 = 0$.

h	2^{-4}	2^{-5}	2^{-6}	2^{-7}	2^{-8}	2^{-9}	2^{-10}
$d = 2$	5	7	12	22	43	83	165
$d = 3$	5	7	10	15	27	-	-

Signorini Problem

Problem Statement

Given $\mathbf{f} : \Omega \rightarrow \mathbb{R}^d$, $\varphi : \Gamma_C \rightarrow \mathbb{R}$, find $\mathbf{u} : \Omega \rightarrow \mathbb{R}^d$, minimizing:

$$\min_{\mathbf{u}} \int_{\Omega} \mu_L |\nabla_S \mathbf{u}|^2 + \frac{\lambda_L}{2} |\operatorname{div} \mathbf{u}|^2 - \mathbf{f} \cdot \mathbf{u} \, dx \quad \text{subject to} \quad \mathbf{u}|_{\Gamma_C} \cdot \mathbf{n} \leq \varphi + \text{bcs.}$$

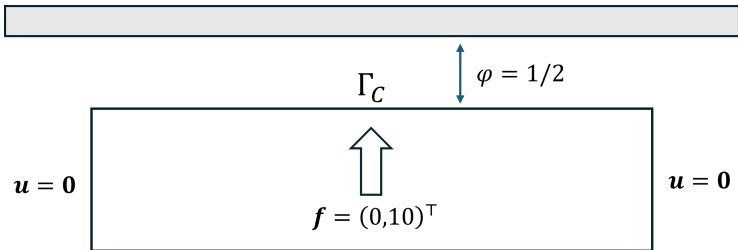
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Pick $\Omega = (0, 5) \times (0, 1)^{d-1}$:



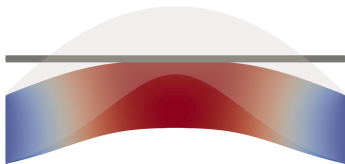
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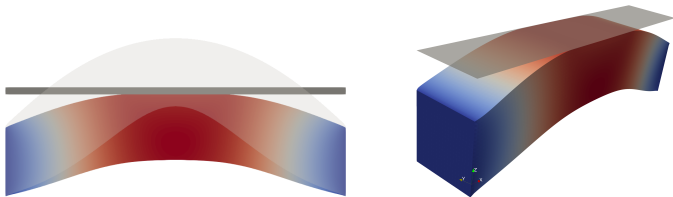
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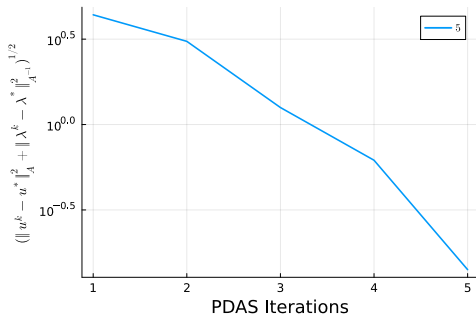
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Signorini Problem Example

h	1/5	1/10	1/20	1/40	1/80	1/160	1/320
$d = 2$	6						
$d = 3$	7						

PDAS iterations to reach convergence with $\mathbf{u}^0 = \mathbf{0}$ with mesh size h .

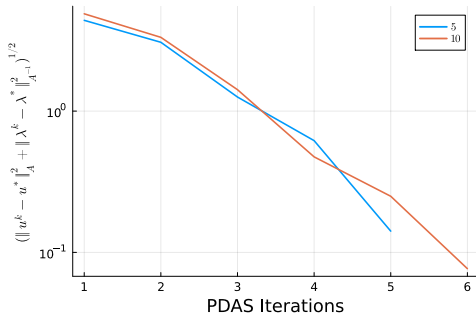


Convergence plot when $d = 2$.

Signorini Problem Example

h	1/5	1/10	1/20	1/40	1/80	1/160	1/320
$d = 2$	6	7					
$d = 3$	7	8					

PDAS iterations to reach convergence with $\mathbf{u}^0 = \mathbf{0}$ with mesh size h .

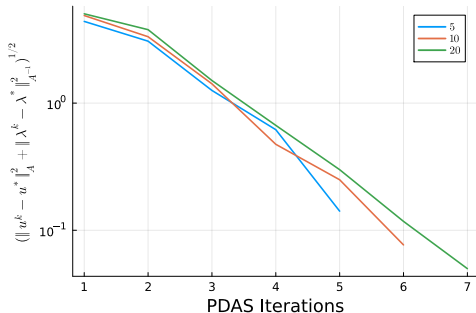


Convergence plot when $d = 2$.

Signorini Problem Example

h	1/5	1/10	1/20	1/40	1/80	1/160	1/320
$d = 2$	6	7	8				
$d = 3$	7	8	9				

PDAS iterations to reach convergence with $\mathbf{u}^0 = \mathbf{0}$ with mesh size h .

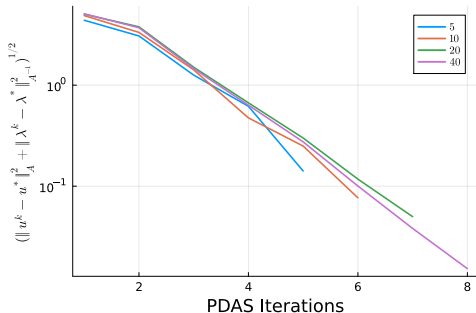


Convergence plot when $d = 2$.

Signorini Problem Example

h	1/5	1/10	1/20	1/40	1/80	1/160	1/320
$d = 2$	6	7	8	9			
$d = 3$	7	8	9	11			

PDAS iterations to reach convergence with $\mathbf{u}^0 = \mathbf{0}$ with mesh size h .

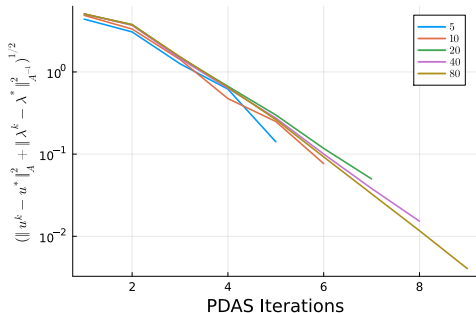


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$d = 2$	6	7	8	9	10		
$d = 3$	7	8	9	11	12		

PDAS iterations to reach convergence with $\mathbf{u}^0 = \mathbf{0}$ with mesh size h .

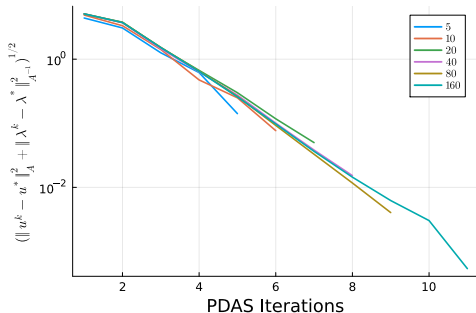


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Signorini Problem Example

h	1/5	1/10	1/20	1/40	1/80	1/160	1/320
$d = 2$	6	7	8	9	10	12	
$d = 3$	7	8	9	11	12	-	

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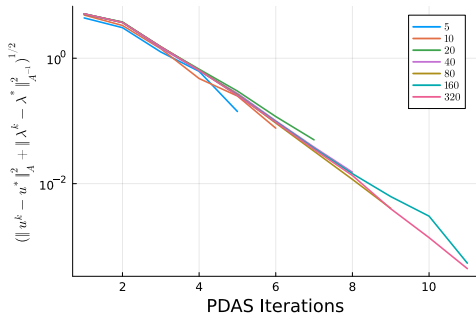


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Signorini Problem Example

h	1/5	1/10	1/20	1/40	1/80	1/160	1/320
$d = 2$	6	7	8	9	10	12	12
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PDAS iterations to reach convergence with $\mathbf{u}^0 = \mathbf{0}$ with mesh size h .



Convergence plot when $d = 2$.

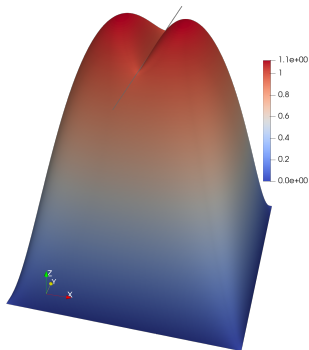
A thin obstacle problem

Consider an obstacle problem with a *thin* obstacle of codimension one. For $(x_1, x_2) = x \in \mathbb{R}^2$,

$$\Omega = (0, 1)^2, \quad f \equiv 20,$$

$$u \in \{v \in H_0^1(\Omega) : v|_{\Gamma_{1/2}} \leq 1\},$$

where $\Gamma_{1/2} := \{x \in \overline{\Omega} : x_1 = 1/2\}$.



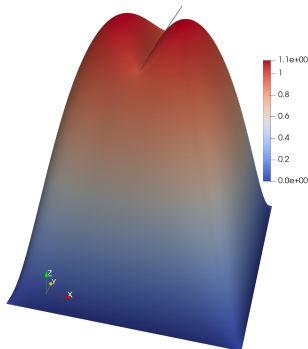
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h	2^{-4}	2^{-5}	2^{-6}	2^{-7}	2^{-8}	2^{-9}	2^{-10}
$d = 2$	3						

PDAS iterations to reach convergence with $\mathbf{u}^0 = \mathbf{0}$ and mesh size h .

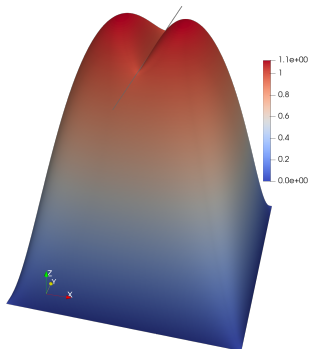
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$d = 2$	3	5					

PDAS iterations to reach convergence with $\mathbf{u}^0 = \mathbf{0}$ and mesh size h .

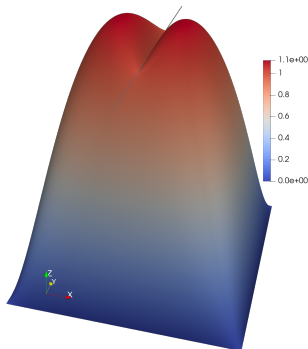
A thin obstacle problem

Consider an obstacle problem with a *thin* obstacle of codimension one. For $(x_1, x_2) = x \in \mathbb{R}^2$,

$$\Omega = (0, 1)^2, \quad f \equiv 20,$$

$$u \in \{v \in H_0^1(\Omega) : v|_{\Gamma_{1/2}} \leq 1\},$$

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h	2^{-4}	2^{-5}	2^{-6}	2^{-7}	2^{-8}	2^{-9}	2^{-10}
$d = 2$	3	5	5				

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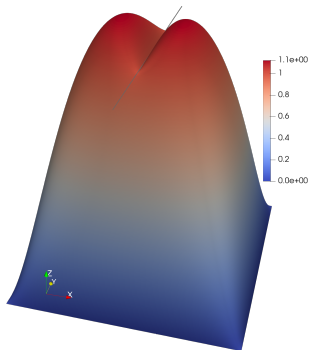
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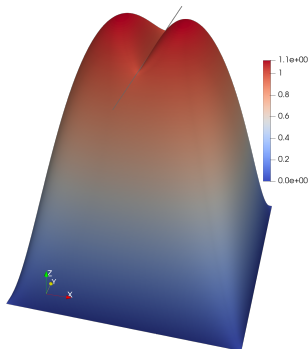
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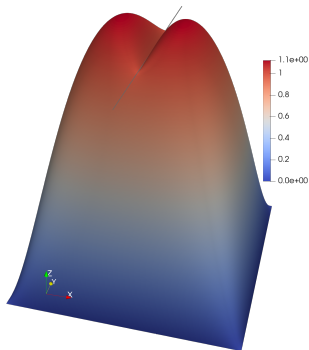
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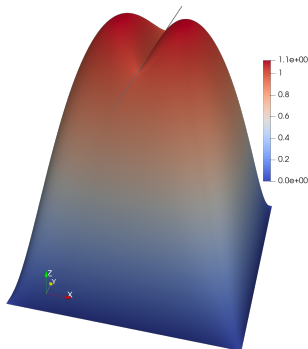
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PDAS iterations to reach convergence with $\mathbf{u}^0 = \mathbf{0}$ and mesh size h .

What is going on?

Obstacle Problem Observations

- Local superlinear convergence.
- The convergence rate seems gets closer to one as $h \rightarrow 0$.
- Iteration counts roughly double with each uniform refinement.

Thin Obstacle/Signorini Problem Observations

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What are primal-dual active set strategies?

Constrained Optimization Problem

$$\min_{\mathbf{u} \in \mathbb{R}^N} \frac{1}{2} \mathbf{u}^\top \mathbf{A} \mathbf{u} - \mathbf{b}^\top \mathbf{u} \quad \text{subject to} \quad \mathbf{u} \leq \boldsymbol{\varphi}. \quad (1)$$

KKT conditions

Suppose $\mathbf{u}^* \in \mathbb{R}^N$ is the minimizer of (1). Then there exists a Lagrange multiplier $\boldsymbol{\lambda}^* \in \mathbb{R}^N$ such that

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$$\mathbf{A} \mathbf{u}^* + \boldsymbol{\lambda}^* = \mathbf{b}, \quad (2)$$

Stationarity

Constrained Optimization Problem

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Stationarity, primal feasibility

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Stationarity, primal feasibility, dual feasibility

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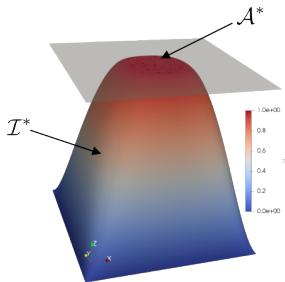
$$\mathbf{A} \mathbf{u}^* + \boldsymbol{\lambda}^* = \mathbf{b}, \quad \mathbf{u}^* \leq \boldsymbol{\varphi}, \quad \boldsymbol{\lambda}^* \geq \mathbf{0}, \quad (\boldsymbol{\lambda}^*)^\top (\mathbf{u}^* - \boldsymbol{\varphi}) = 0. \quad (2)$$

Stationarity, primal feasibility, dual feasibility and complementarity.

The active & inactive sets

The active set is the region where the membrane is pushing against the obstacle:

$$\mathcal{A}^* = \{i : \mathbf{u}_i^* = \varphi_i \text{ and } \lambda_i^* > 0\} \text{ and conversely } \mathcal{I}^* = \{i : \mathbf{u}_i^* \leq \varphi_i \text{ and } \lambda_i^* = 0\}.$$

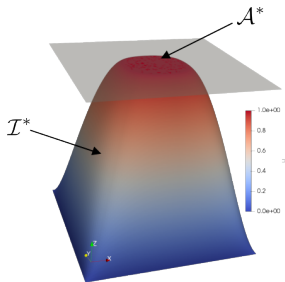


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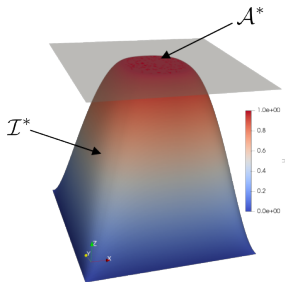
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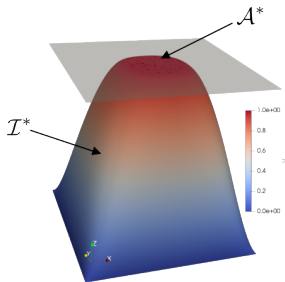
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Each iteration of a PDAS makes a guess of the true active/inactive sets and fixes $\mathbf{u}^{k+1}$ to the obstacle in the estimated active set and solves the Poisson problem in the estimated inactive set.

Hintermüller–Ito–Kunisch (HIK) PDAS (2002)

Given the iterates $\mathbf{u}^k$ and $\boldsymbol{\lambda}^k$ define:

- Active set iterate $\mathcal{A}^k = \{i : \boldsymbol{\lambda}_i^k + \mathbf{u}_i^k - \varphi_i > 0\}$,
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Compute the inactive set update for $\mathbf{u}^k$ by solving:

$$A_{\mathcal{I}^k, \mathcal{I}^k} \delta_{\mathcal{I}^k}^k = -A_{\mathcal{I}^k, \mathcal{A}^k} (\varphi - \mathbf{u}^k)_{\mathcal{A}^k} - (A\mathbf{u}^k - \mathbf{b})_{\mathcal{I}^k},$$

Then update:

$$\mathbf{u}_{\mathcal{I}^k}^{k+1} = \mathbf{u}_{\mathcal{I}^k}^k + \delta_{\mathcal{I}^k}^k, \quad \mathbf{u}_{\mathcal{A}^k}^{k+1} = \varphi_{\mathcal{A}^k}, \quad \boldsymbol{\lambda}_{\mathcal{A}^k}^{k+1} = -(A\mathbf{u}^{k+1} - \mathbf{b})_{\mathcal{A}^k} \quad \text{and} \quad \boldsymbol{\lambda}_{\mathcal{I}^k}^{k+1} = \mathbf{0}.$$

A novel convergence result

$$\|\mathbf{u}^{k+1} - \mathbf{u}^*\|_A = \left(1 - \frac{\|\boldsymbol{\lambda}_{\mathcal{I}^k}^k\|_{(A_{\mathcal{I}^k, \mathcal{I}^k})^{-1}}^2}{\|\boldsymbol{\lambda}^k - \boldsymbol{\lambda}^*\|_{A^{-1}}^2} \right)^{1/2} \|\mathbf{u}^k - \mathbf{u}^*\|_A.$$

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Ito, Kunisch (2003)

...an essential drawback of active set strategies [...] changes from one iteration to the next occur along layers between active and inactive sets which have only the width of one mesh-size. For fine mesh-sizes this results in large iteration numbers.

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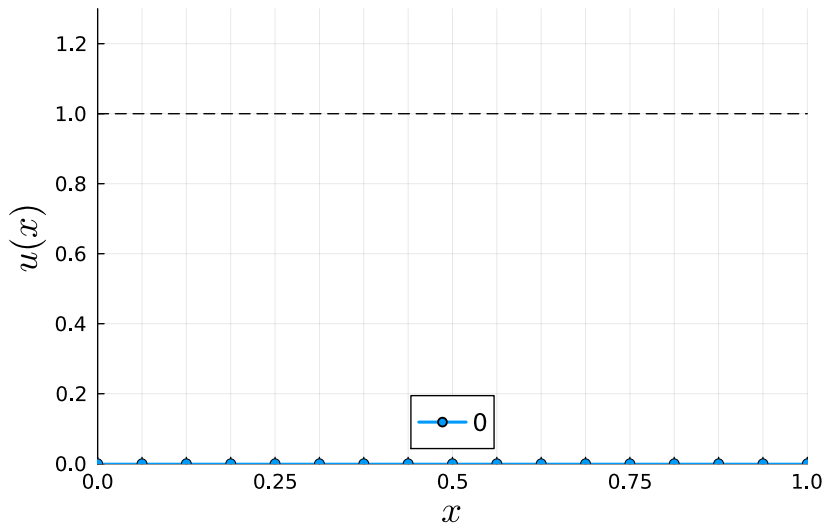
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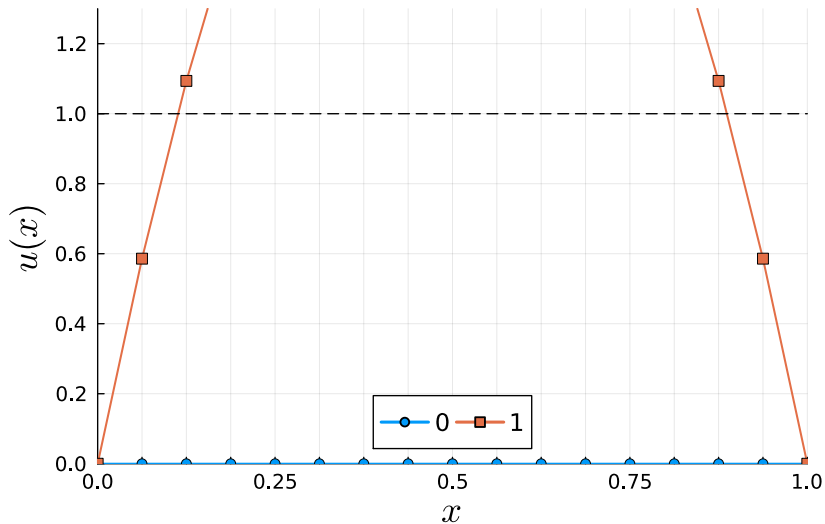
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Layer-by-layer peeling

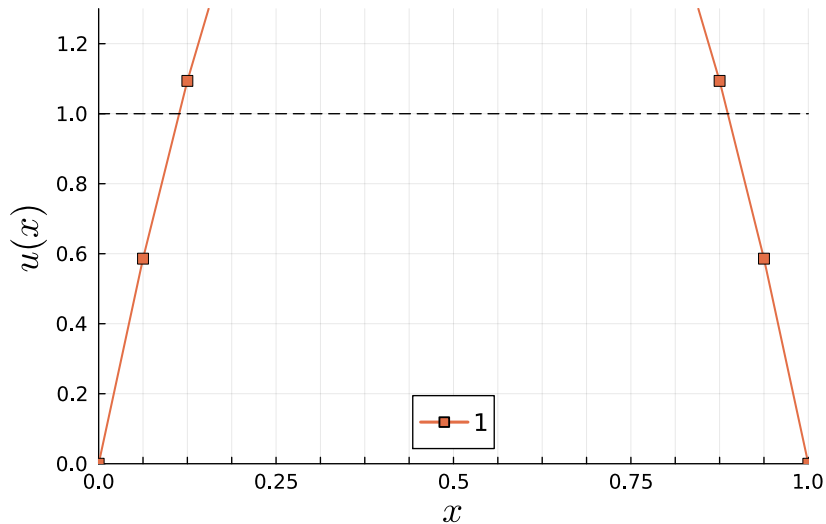
Obstacle Problem Example Revisited



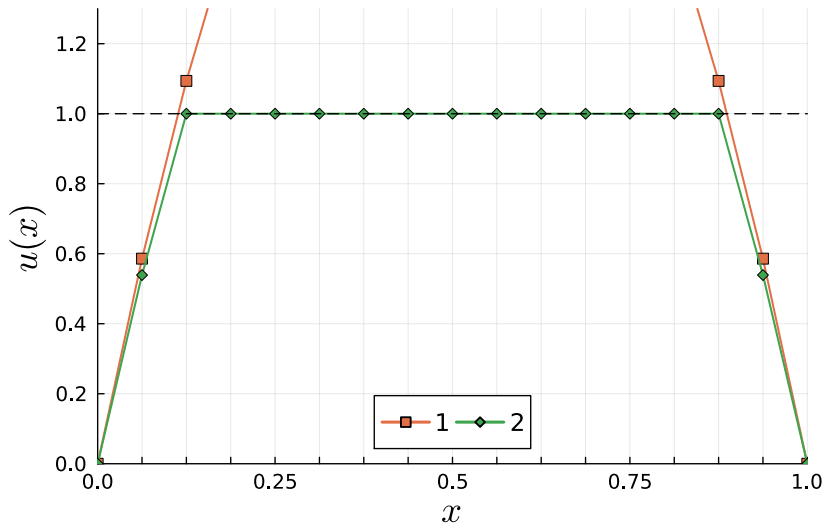
Obstacle Problem Example Revisited



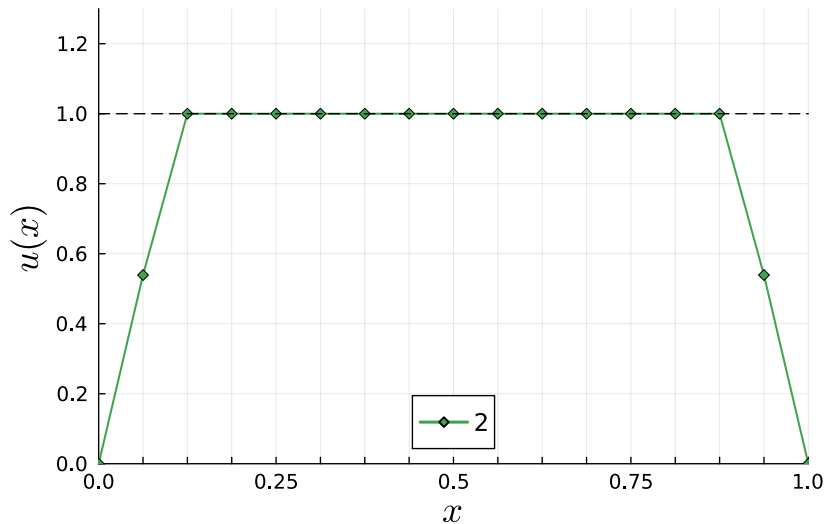
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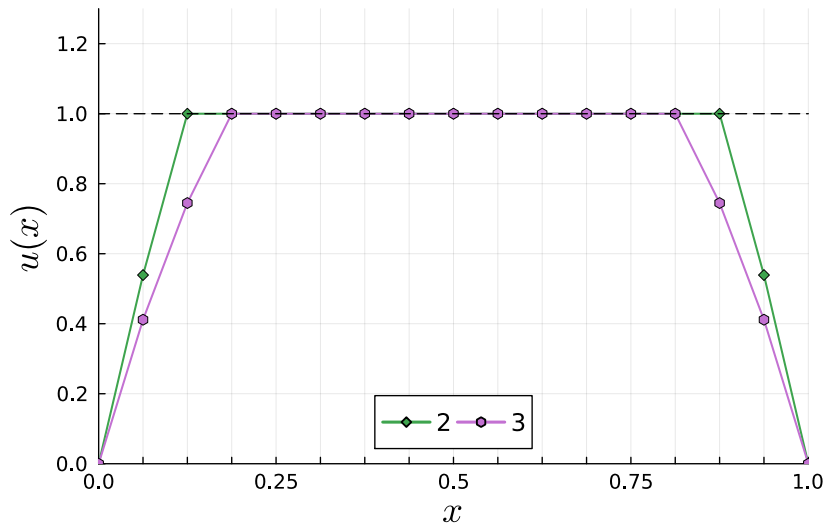
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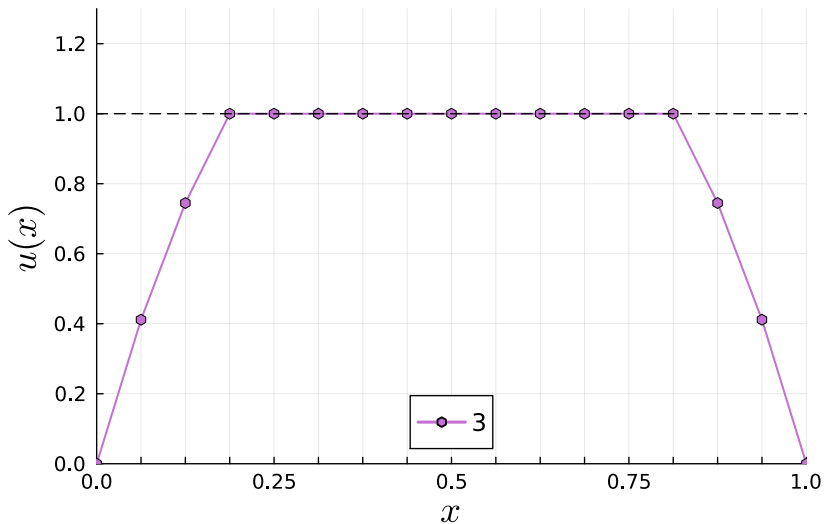
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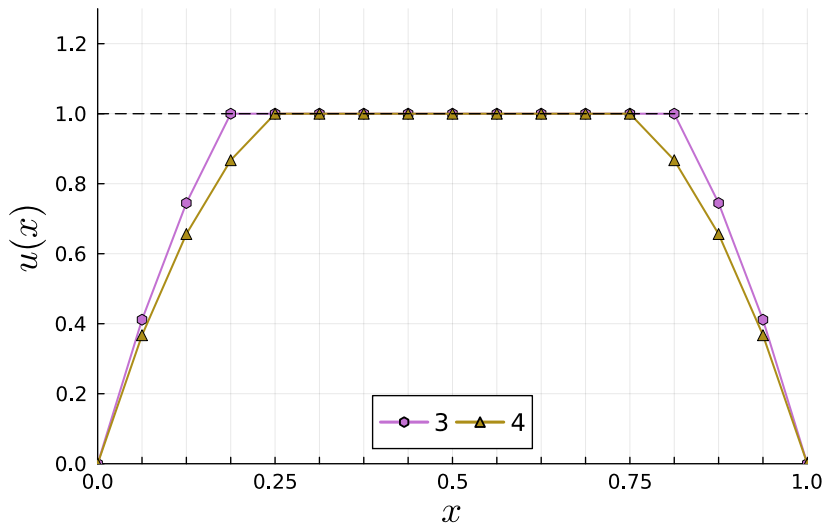
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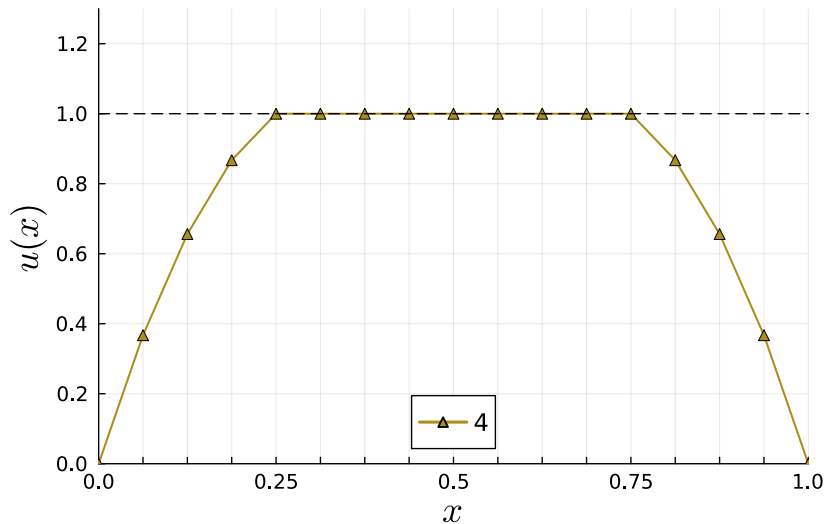
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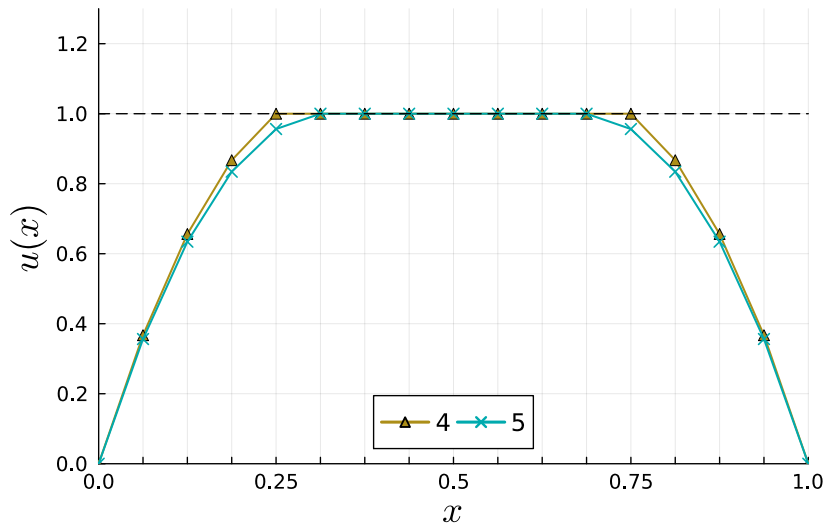
Obstacle Problem Example Revisited



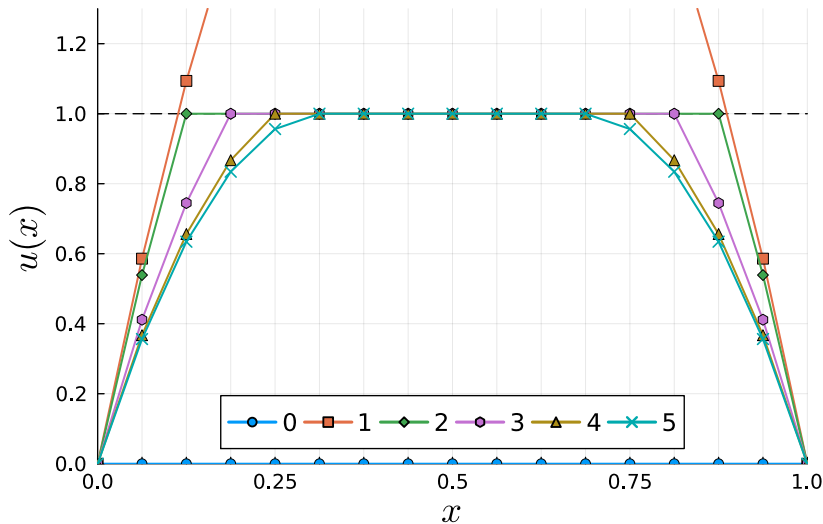
Obstacle Problem Example Revisited



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Obstacle Problem Example Revisited

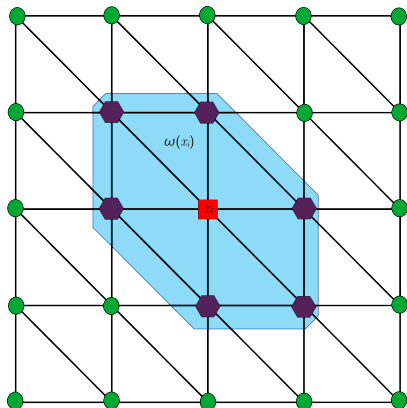


Layer-by-layer peeling

► Play video

Peeling Theorem

Suppose a dof lives in the interior of the active set $\mathcal{A}^k$ at the k^{th} HIK-PDAS iteration. Then the same dof also lives in the active set $\mathcal{A}^{k+1}$ at the $(k+1)^{\text{th}}$ HIK-PDAS iteration.

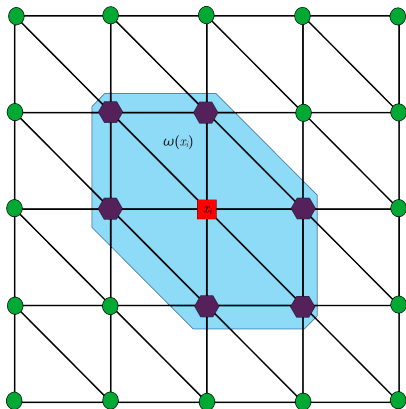


Peeling Corollary

Only dofs on the boundary of the current active set $\mathcal{A}^k$ can switch to the inactive set in the next iteration.

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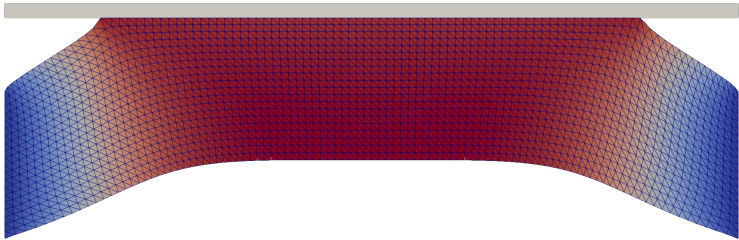
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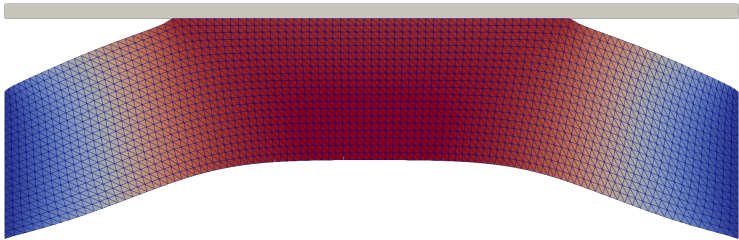
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Signorini Problem Example Revisited



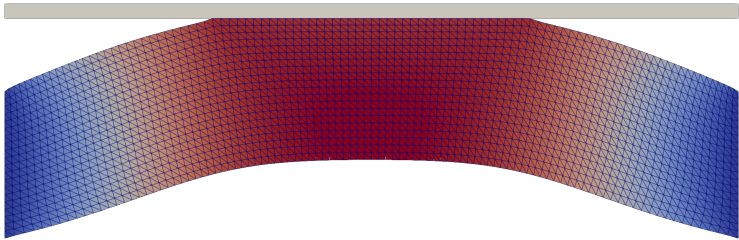
PDAS Iterate 2: 69 active dofs

Signorini Problem Example Revisited



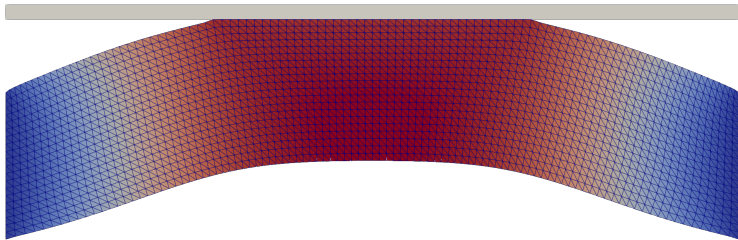
PDAS Iterate 3: 51 active dofs

Signorini Problem Example Revisited



PDAS Iterate 4: 41 active dofs

Signorini Problem Example Revisited



PDAS Iterate 4: 41 active dofs

Active dofs on the top boundary *always* neighbour an “inactive” dof in the bulk.

Semismooth Newton methods

Newton methods for optimization

Consider the unconstrained problem $\min_u J(u)$. Define $F(u) = J'(u)$ as the Fréchet derivative of J . Under sufficient conditions, if u^* is a minimizer then

$$F(u^*) = 0.$$

Newton's method

Given an iterate u^k , one seeks a Newton update δ^k that satisfies:

$$F'(u^k)\delta^k = -F(u^k)$$

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Convergence Theorem

If F is locally Lipschitz continuous and Fréchet differentiable at a solution u^* and $F'(u^*)$ is nonsingular, then there exists a neighbourhood $B(u^*)$ centred at u^* such that if $u^k \in B(u^*)$ then

$$\|u^{k+1} - u^*\| \leq c_k \|u^k - u^*\|$$

for $0 \leq c_k < 1$ and $c_k \rightarrow 0$ as $k \rightarrow \infty$. This is called *local superlinear convergence*.

Can a similar solver be designed for inequality constrained problems?

Yes! They require a generalization called *semismooth* Newton methods.

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KKT as nonsmooth system of equations

How do we deal with the inequalities in KKT?

Nonsmooth reformulation of KKT

The KKT conditions can be rewritten as

$$\begin{aligned} Au + \lambda &= b, \\ \lambda - (\lambda + u - \varphi)_+ &= 0, \end{aligned} \tag{3}$$

where, for $\mathbf{a} \in \mathbb{R}^N$, $[(\mathbf{a})_+]_i = \max(\mathbf{a}_i, 0)$.

Good News

(3) is a system of equations.

Bad News

The system of equations is *nonsmooth*. The $(\cdot)_+$ operator does not possess a Fréchet derivative and (classical) Newton's method is ill-defined.

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Semismooth Newton methods

A critical observation

The $(\cdot)_+$ operator possesses a *Newton* derivative.

Newton derivative

Let V and W be Banach spaces with $U \subset V$ open. We say that the $G : U \rightarrow \mathcal{L}(V, W)$ is the Newton derivative of $F : U \rightarrow W$ if

$$\lim_{\|\epsilon\|_V \rightarrow 0} \frac{\|F(u + \epsilon) - F(u) - G(u + \epsilon)(\epsilon)\|_W}{\|\epsilon\|_V} = 0.$$

Theorem

The following operator G_+ is a Newton derivative for $(\mathbf{v})_+$:

$$[G_+(\mathbf{v})\mathbf{w}]_i = \begin{cases} \mathbf{w}_i & \text{if } \mathbf{v}_i > 0, \\ 0 & \text{otherwise.} \end{cases}$$

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Semismooth Newton methods

Substituting the Newton derivative in place of the Fréchet derivative in Newton's method is sufficient to recover local superlinear convergence. These are known as *semismooth* Newton methods (SSN).

An SSN for (3)

Let $\mathcal{A}^k = \{i : \lambda_i^k + \mathbf{u}_i^k - \varphi_i > 0\}$ and $\mathcal{I}^k = \{i : \lambda_i^k + \mathbf{u}_i^k - \varphi_i \leq 0\}$.

Given the SSN iterates $(\mathbf{u}^k, \lambda^k)$, compute the SSN updates (δ^k, ϵ^k) satisfying:

$$\begin{pmatrix} A & I_N \\ -E^{\mathcal{A}^k} & E^{\mathcal{I}^k} \end{pmatrix} \begin{pmatrix} \delta^k \\ \epsilon^k \end{pmatrix} = - \begin{pmatrix} A\mathbf{u}^k + \lambda^k - \mathbf{b} \\ \mathbf{g}^k \end{pmatrix}, \quad (4)$$

where I_N is the $N \times N$ identity matrix and for $\mathcal{K} \subseteq \{1:N\}$,

$$[E^{\mathcal{K}}]_{ij} = \begin{cases} 1 & \text{if } i = j \in \mathcal{K}, \\ 0 & \text{otherwise,} \end{cases} \quad \mathbf{g}_i^k = \begin{cases} \varphi_i - \mathbf{u}_i^k & \text{if } i \in \mathcal{A}^k, \\ \lambda_i^k & \text{if } i \in \mathcal{I}^k. \end{cases}$$

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SSN $\iff$ HIK-PDAS

Some linear algebra reveals that the SSN linear system can be decomposed into:

$$A_{\mathcal{I}^k, \mathcal{I}^k} \delta_{\mathcal{I}^k}^k = -A_{\mathcal{I}^k, \mathcal{A}^k} (\varphi - \mathbf{u}^k)_{\mathcal{A}^k} - (A\mathbf{u}^k - \mathbf{b})_{\mathcal{I}^k},$$

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This is the same as the HIK-PDAS, i.e. the SSN and HIK-PDAS steps are equivalent.

Local superlinear convergence

Since the SSN converges with a local superlinear rate and HIK-PDAS is equivalent to SSN, then HIK-PDAS converges at a local superlinear rate.

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What does the literature say? Regularity

Ito, Kunisch (2003)

The fact that the Lagrange multipliers [iterates] of the auxiliary problems in [...] the algorithm are not contained in the pivot space $L^2(\Omega)$ between $H_0^1(\Omega)$ and $H^{-1}(\Omega)$ presents a serious difficulty.

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At this point we remark that the primal-dual active set strategy has no straightforward infinite dimensional analogue for [...] obstacle problems [...] as the Lagrange multiplier is only a measure in general, and hence the core steps [...] of our algorithm are no longer meaningful.

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Infinite-dimensional Obstacle Problem

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Under sufficient regularity conditions (e.g. convex domain), there exists a Lagrange multiplier $\lambda^* \in L^2(\Omega)$ such that the minimizer $u^* \in H_0^1(\Omega)$ of the obstacle problem satisfies: for all $v \in H_0^1(\Omega)$

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Norm gap (Ulbrich, 2002)

$(\cdot)_+ : L^q(\Omega) \rightarrow L^p(\Omega)$ possesses a Newton derivative if and only if $p < q$.

Consequence

There is no infinite-dimensional analogue of the SSN for the obstacle problem. $G_+ : L^q(\Omega) \rightarrow \mathcal{L}(L^q(\Omega), L^p(\Omega))$ is only invertible when $p = q$ but then it is not a Newton derivative.

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Infinite-dimensional Obstacle Problem

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Under sufficient regularity conditions (e.g. convex domain), there exists a Lagrange multiplier $\lambda^* \in L^2(\Omega)$ such that the minimizer $u^* \in H_0^1(\Omega)$ of the obstacle problem satisfies: for all $v \in H_0^1(\Omega)$

$$\begin{aligned}(\nabla u^*, \nabla v)_{L^2(\Omega)} + (\lambda^*, v)_{L^2(\Omega)} &= (f, v)_{L^2(\Omega)}, \\ \lambda^* - (\lambda^* + u^* - \varphi)_+ &= 0 \text{ a.e. in } \Omega.\end{aligned}$$

Norm gap (Ulbrich, 2002)

$(\cdot)_+ : L^q(\Omega) \rightarrow L^p(\Omega)$ possesses a Newton derivative if and only if $p < q$.

Consequence

There is no infinite-dimensional analogue of the SSN for the obstacle problem. $G_+ : L^q(\Omega) \rightarrow \mathcal{L}(L^q(\Omega), L^p(\Omega))$ is only invertible when $p = q$ but then it is not a Newton derivative.

This explains the loss of local superlinear convergence of PDAS when $h \rightarrow 0$ for the obstacle and Signorini problems.

Obstacle Problem and Ill-Posed HIK-PDAS

Infinite-dimensional HIK-PDAS

Suppose we are given $u^k, \lambda^k \in L^1(\Omega)$. Define

$$\mathcal{A}^k = \{\lambda^k + u^k - \varphi > 0 \text{ a.e.}\} \text{ and } \mathcal{I}^k = \text{int}(\Omega \setminus \mathcal{A}^k).$$

Then the HIK-PDAS iteration is to find $u^{k+1} \in H_0^1(\Omega)$ satisfying

$$\begin{aligned}(\nabla u^{k+1}, \nabla v) &= (f, v) \quad \forall v \in H_0^1(\mathcal{I}^k), \\ u^{k+1}|_{\mathcal{A}^k} &= \varphi_{\mathcal{A}^k},\end{aligned}$$

Fix $\langle \lambda^{k+1}, w \rangle = (f, w) - (\nabla u^{k+1}, \nabla w) \quad \forall w \in H_0^1(\Omega)$.

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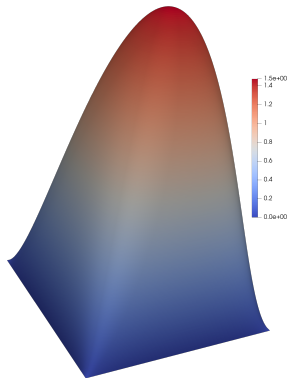
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Obstacle Problem and Ill-Posed HIK-PDAS

Pick $\Omega = (0, 1)^2$, $f = 20$, $\varphi = 1$, $u^0 = \lambda^0 = 0$.



(a) u^1

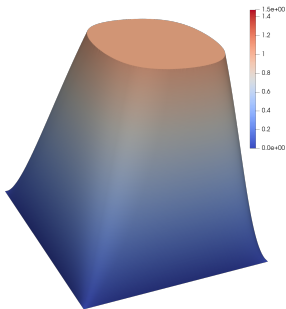


(b) λ^1

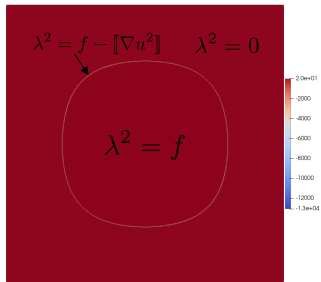
$$\mathcal{A}^1 = \{u^1 > \varphi\}.$$

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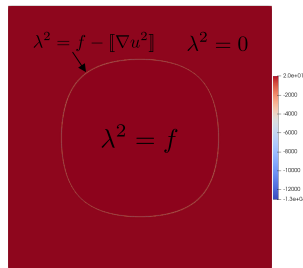


(a) u^2



(b) λ^2

Obstacle Problem and Ill-Posed HIK-PDAS



$$\lambda^2 = f|_{\mathcal{A}^1} - \llbracket \nabla u^2 \rrbracket \delta_{\partial \mathcal{A}^1}$$

where $\llbracket \cdot \rrbracket$ denotes the jump across the boundary of $\mathcal{A}^1$ and δ is the Delta function of codimension one.

λ^2 has no pointwise a.e. interpretation and $\mathcal{A}^2$ is ill-defined.

This happens even though $\lambda^* \in L^2(\Omega)$.

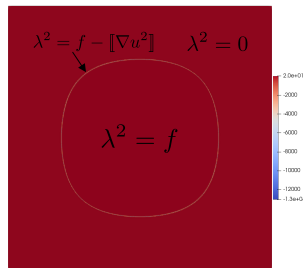
Deterioration of convergence rate

Recall that (in finite dimensions) $\|\lambda_{\mathcal{I}^k}^k\|_{(A_{\mathcal{I}^k, \mathcal{I}^k})^{-1}}^2$ determines the rate of convergence.

[Conjecture] $\lim_{h \rightarrow 0} \|\lambda_{\mathcal{I}^2}^2\|_{(A_{\mathcal{I}^2, \mathcal{I}^2})^{-1}}^2 = \|\lambda^2\|_{H^{-1}(\mathcal{I}^1)}^2$.

But $\|\lambda^2\|_{H^{-1}(\mathcal{I}^1)} = 0$ hence the rate deteriorates to one.

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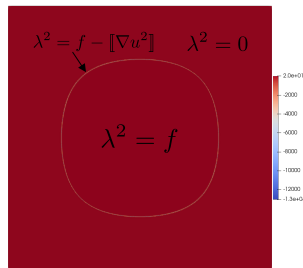
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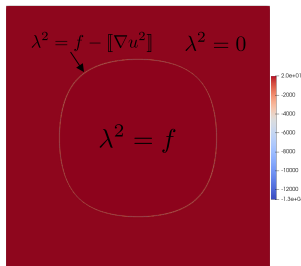
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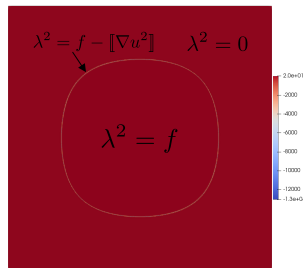
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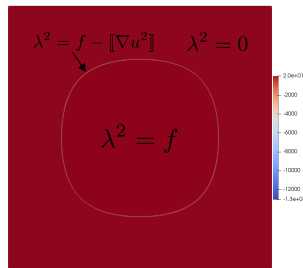
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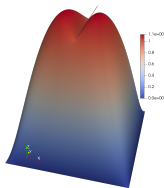
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Thin Obstacle Problem and Well-Posed HIK-PDAS



(a) u^*

Regularity on $\Gamma_{1/2} = \{x_1 = 1/2\}$

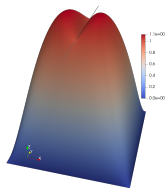
$\lambda^2 \in L^q(\Gamma_{1/2})$ for $q \in [1, 8/5)$. $\mathcal{A}^2 = \{[\![\nabla u^2]\!] < 0 \text{ a.e. on } \Gamma_{1/2}\}$ is well-defined.

Convergence Theorem

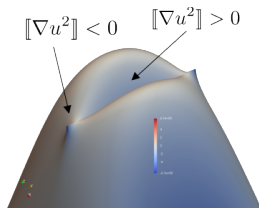
HIK-PDAS converges at the following guaranteed sublinear rate:

$$\|\nabla(u^{k+1} - u^*)\|_{L^2(\Omega)} = \left(1 - \frac{\|\lambda^k\|_{H^{-1}(\mathcal{I}^k)}^2}{\|\lambda^k - \lambda^*\|_{H^{-1}(\Omega)}^2}\right)^{1/2} \|\nabla(u^k - u^*)\|_{L^2(\Omega)}.$$

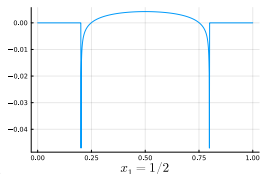
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(a) u^*



(b) u^2



(c) λ^2

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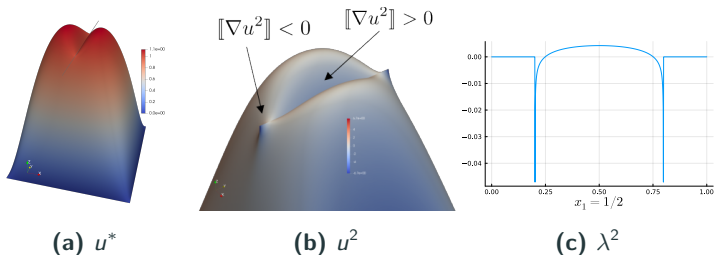
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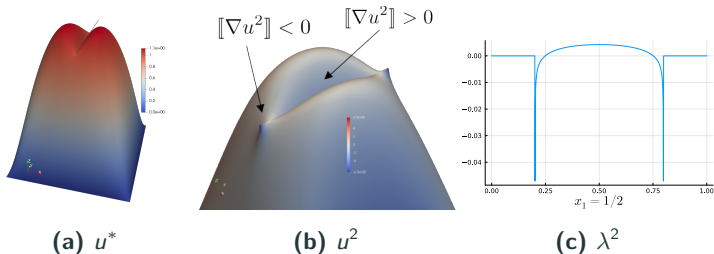
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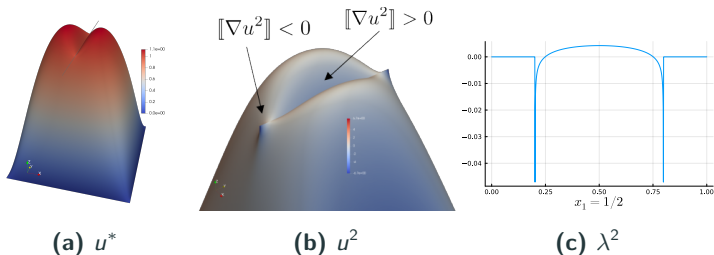
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Two remedies

One can *regularize* the KKT nonsmooth system, e.g. for all $v \in H_0^1(\Omega)$, and some $\alpha \in (0, 1)$,

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This system now possesses a Newton derivative and a PDAS can be developed. But we need to send $\alpha \rightarrow 1$ to recover the original solution.

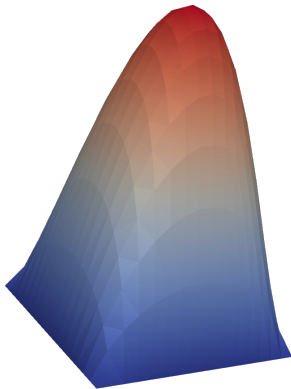
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Grid-sequencing/multigrid

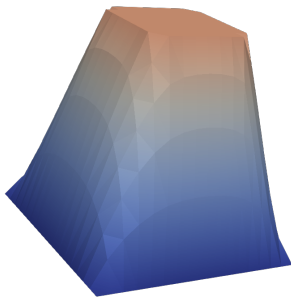
h	2^{-4}	2^{-5}	2^{-6}	2^{-7}	2^{-8}	2^{-9}	2^{-10}
PDAS iterations							



$h = 2^{-4}$: PDAS iteration 1.

Grid-sequencing/multigrid

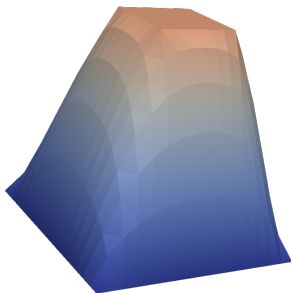
h	2^{-4}	2^{-5}	2^{-6}	2^{-7}	2^{-8}	2^{-9}	2^{-10}
PDAS iterations							



$h = 2^{-4}$: PDAS iteration 2.

Grid-sequencing/multigrid

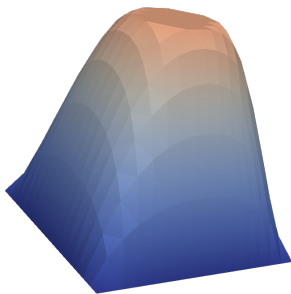
h	2^{-4}	2^{-5}	2^{-6}	2^{-7}	2^{-8}	2^{-9}	2^{-10}
PDAS iterations							



$h = 2^{-4}$: PDAS iteration 3.

Grid-sequencing/multigrid

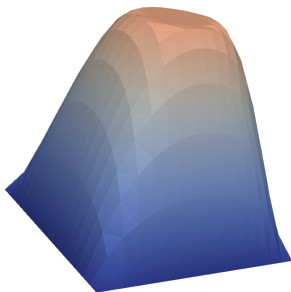
h	2^{-4}	2^{-5}	2^{-6}	2^{-7}	2^{-8}	2^{-9}	2^{-10}
PDAS iterations							



$h = 2^{-4}$: PDAS iteration 4.

Grid-sequencing/multigrid

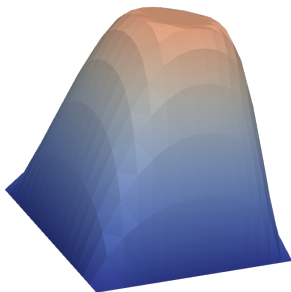
h	2^{-4}	2^{-5}	2^{-6}	2^{-7}	2^{-8}	2^{-9}	2^{-10}
PDAS iterations	5						



$h = 2^{-4}$: PDAS iteration 5.

Grid-sequencing/multigrid

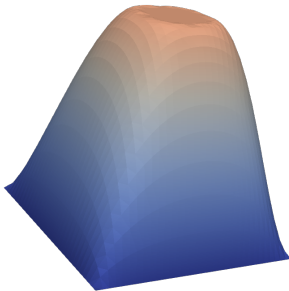
h	2^{-4}	2^{-5}	2^{-6}	2^{-7}	2^{-8}	2^{-9}	2^{-10}
PDAS iterations	5						



$h = 2^{-5}$: PDAS iteration 0.

Grid-sequencing/multigrid

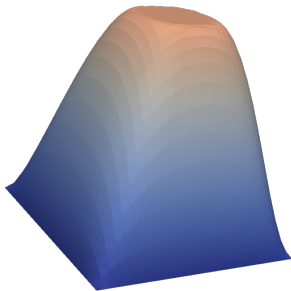
h	2^{-4}	2^{-5}	2^{-6}	2^{-7}	2^{-8}	2^{-9}	2^{-10}
PDAS iterations	5						



$h = 2^{-5}$: PDAS iteration 1.

Grid-sequencing/multigrid

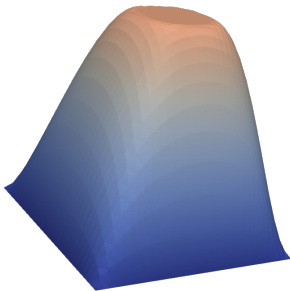
h	2^{-4}	2^{-5}	2^{-6}	2^{-7}	2^{-8}	2^{-9}	2^{-10}
PDAS iterations	5	2					



$h = 2^{-5}$: PDAS iteration 2.

Grid-sequencing/multigrid

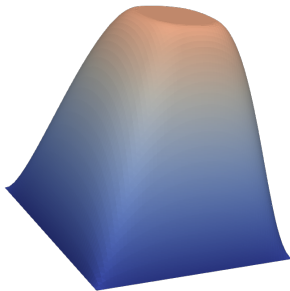
h	2^{-4}	2^{-5}	2^{-6}	2^{-7}	2^{-8}	2^{-9}	2^{-10}
PDAS iterations	5	2					



$h = 2^{-6}$: PDAS iteration 0.

Grid-sequencing/multigrid

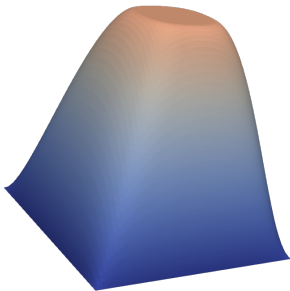
h	2^{-4}	2^{-5}	2^{-6}	2^{-7}	2^{-8}	2^{-9}	2^{-10}
PDAS iterations	5	2					



$h = 2^{-6}$: PDAS iteration 1.

Grid-sequencing/multigrid

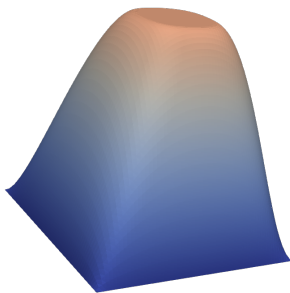
h	2^{-4}	2^{-5}	2^{-6}	2^{-7}	2^{-8}	2^{-9}	2^{-10}
PDAS iterations	5	2	2				



$h = 2^{-6}$: PDAS iteration 2.

Grid-sequencing/multigrid

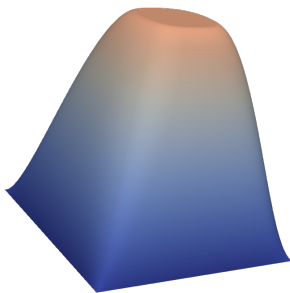
h	2^{-4}	2^{-5}	2^{-6}	2^{-7}	2^{-8}	2^{-9}	2^{-10}
PDAS iterations	5	2	2				



$h = 2^{-7}$: PDAS iteration 0.

Grid-sequencing/multigrid

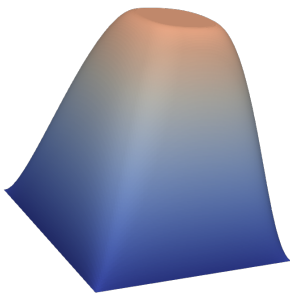
h	2^{-4}	2^{-5}	2^{-6}	2^{-7}	2^{-8}	2^{-9}	2^{-10}
PDAS iterations	5	2	2				



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Grid-sequencing/multigrid

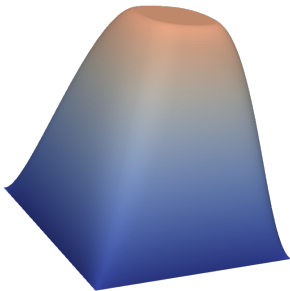
h	2^{-4}	2^{-5}	2^{-6}	2^{-7}	2^{-8}	2^{-9}	2^{-10}
PDAS iterations	5	2	2				



$h = 2^{-7}$: PDAS iteration 2.

Grid-sequencing/multigrid

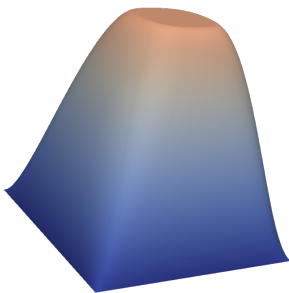
h	2^{-4}	2^{-5}	2^{-6}	2^{-7}	2^{-8}	2^{-9}	2^{-10}
PDAS iterations	5	2	2	3			



$h = 2^{-7}$: PDAS iteration 3.

Grid-sequencing/multigrid

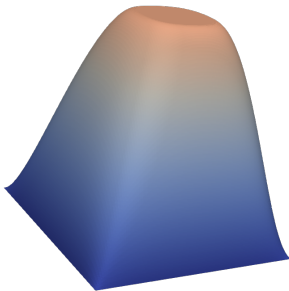
h	2^{-4}	2^{-5}	2^{-6}	2^{-7}	2^{-8}	2^{-9}	2^{-10}
PDAS iterations	5	2	2	3	3		



$h = 2^{-7}$: PDAS iteration 3.

Grid-sequencing/multigrid

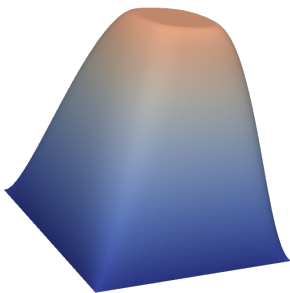
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$h = 2^{-7}$: PDAS iteration 3.

Grid-sequencing/multigrid

h	2^{-4}	2^{-5}	2^{-6}	2^{-7}	2^{-8}	2^{-9}	2^{-10}
PDAS iterations	5	2	2	3	3	3	4



$h = 2^{-7}$: PDAS iteration 3.

Conclusions

- Primal-dual active set strategies (PDAS) are equivalent to semismooth Newton methods **in finite dimensions**.
- The equivalence can be lost in infinite dimensions. This leads to mesh-dependent iteration counts.
- For infinite-dimensional thin obstacle/Signorini problems, PDAS still converge but at a linear rate.
- For infinite-dimensional obstacle problems, PDAS stagnate.

Mesh-dependent iteration count growth in primal-dual active set strategies

I. P., M. Hintermüller, <http://arxiv.org/abs/2607.26622> (2026).

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Thank you for listening!

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